

Exponential and Logarithmic Functions

ECON 320 - Introduction to Mathematical Economics

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1 The Nature of Exponential Functions

Simple exponential functions

Definition 1 (Exponential function). A function is *exponential* if the variable appears in the exponent: $f(t) = a^t$ with base $a > 0, a \neq 1$. The natural-base form uses $a = e$: $f(t) = e^t$.

Key features for $a > 1$: strictly increasing, $f(0) = 1$, and $\lim_{t \rightarrow \infty} f(t) = \infty$. For $0 < a < 1$: strictly decreasing and $\lim_{t \rightarrow \infty} f(t) = 0$.

Graphical form & growth intuition

Exponential growth is *proportional* growth: the instantaneous percentage change is constant (formalized in §10.2). This distinguishes it from polynomials, whose *absolute* change tends to dominate at large t .

Generalized exponential function

Definition 2 (Generalized form). $F(t) = A e^{rt}$ with scale parameter $A > 0$ and constant growth rate $r \in \mathbb{R}$.

If $r > 0$ we have growth; if $r < 0$ decay.

Example 1 (Comparing bases). Show $2^t = e^{(\ln 2)t}$. *Solution:* since $a = e^{\ln a}$, $2^t = (e^{\ln 2})^t = e^{(\ln 2)t}$. Thus all bases can be written in natural-base form with growth rate $r = \ln a$.

A preferred base

The base $e \approx 2.71828$ simplifies differentiation/integration (§10.5) and connects cleanly to continuous compounding/growth (§10.2).

2 Natural Exponential Functions and the Problem of Growth

The number e and continuous compounding

$$\left(1 + \frac{r}{n}\right)^{nt} \xrightarrow{n \rightarrow \infty} e^{rt}.$$

If a principal A_0 is compounded continuously at rate r , then $A(t) = A_0 e^{rt}$.

Instantaneous (proportional) growth rate

Definition 3 (Growth rate). For differentiable $Q(t) > 0$, the instantaneous *proportional* growth rate is

$$g_Q(t) \equiv \frac{1}{Q(t)} \frac{dQ}{dt} = \frac{d}{dt} \ln Q(t).$$

If $Q(t) = A e^{rt}$, then $g_Q(t) = r$ (constant). Thus exponential functions encode constant percentage growth.

Example 2 (Doubling time; Rule of 70). For $Q(t) = A e^{rt}$, the doubling time T_2 solves $2 = A e^{rT_2} / A \Rightarrow T_2 = \frac{\ln 2}{r} \approx \frac{0.693}{r}$. For small r in percent, $T_2 \approx \frac{70}{100r\%}$.

Continuous vs. discrete growth

Discrete compounding at per-period rate R : $Q_t = A(1 + R)^t$. Continuous compounding with rate r : $Q(t) = A e^{rt}$. They coincide if $r = \ln(1 + R)$.

Discounting and negative growth

With continuous discount rate $r > 0$, the present value (PV) of a payoff X at time t is $PV = e^{-rt} X$. This is exponential *decay*.

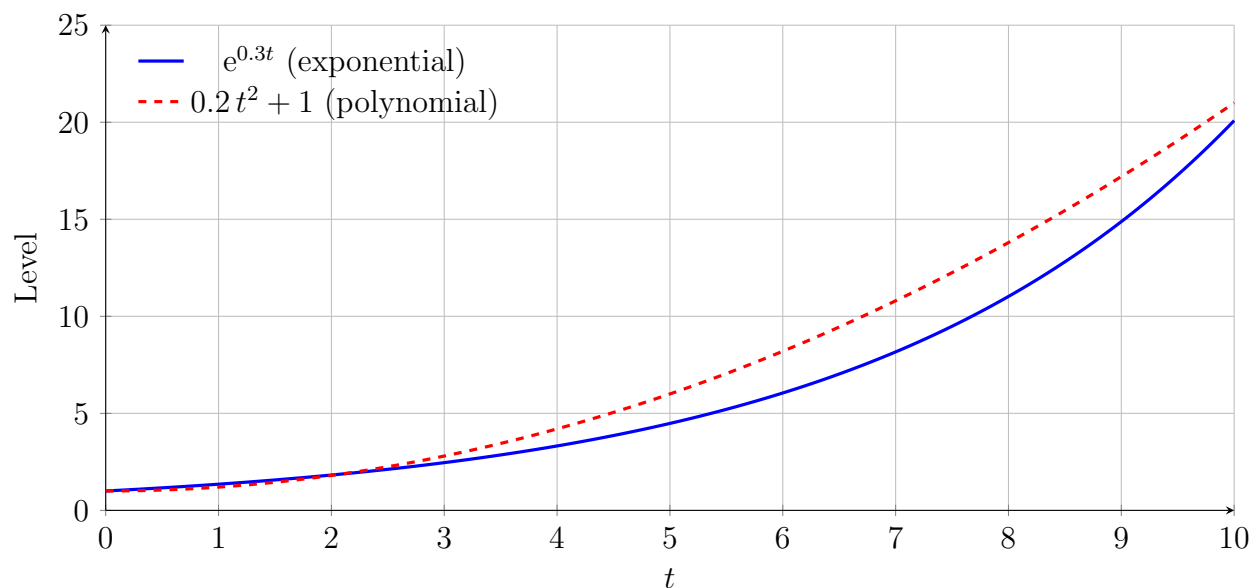


Figure 1: Exponential (constant percentage growth) vs. polynomial (accelerating absolute growth).

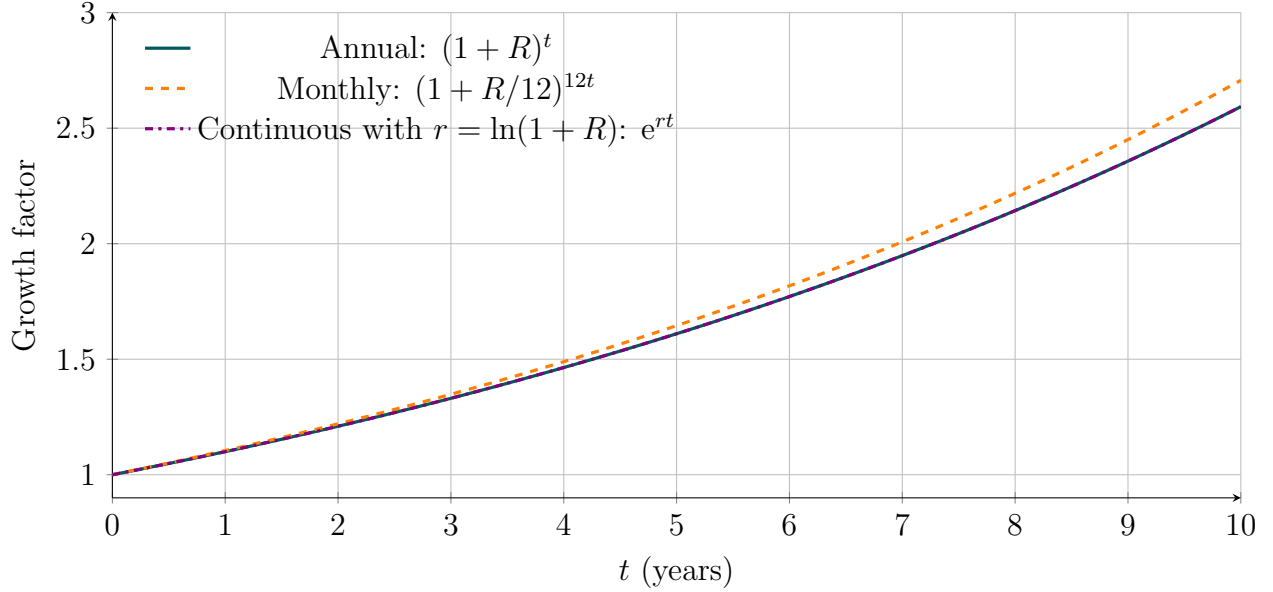


Figure 2: Comparing compounding conventions for $R = 10\%$. Monthly compounding dominates annual; continuous with $r = \ln(1 + R)$ coincides with annual $(1 + R)^t$.

3 Logarithms

Meaning; common vs. natural logs

Definition 4 (Logarithm). For $a > 0, a \neq 1$, $\log_a x = y$ iff $a^y = x$. Natural log $\ln x \equiv \log_e x$; common log $\log x \equiv \log_{10} x$.

Rules of logarithms

For $x, y > 0$ and constant k :

$$\ln(xy) = \ln x + \ln y, \quad \ln\left(\frac{x}{y}\right) = \ln x - \ln y, \quad \ln x^k = k \ln x.$$

Change of base: $\log_a x = \frac{\ln x}{\ln a}$.

Example 3 (Linearizing multiplicative models). Cobb–Douglas output $Y = AK^\alpha L^\beta$ becomes

$$\ln Y = \ln A + \alpha \ln K + \beta \ln L,$$

a linear relation in $\ln K, \ln L$. Coefficients α, β are elasticities of Y w.r.t. K, L (see §10.7).

4 Logarithmic Functions

Log functions as inverses of exponentials

$\ln(e^x) = x$ and $e^{\ln x} = x$ for $x > 0$. Hence $\ln$ is strictly increasing on $(0, \infty)$ with range $\mathbb{R}$.

Graphical form and base conversion

$\ln x$ is concave, with vertical asymptote at $x = 0^+$ and slope $1/x$. Any base a gives $\log_a x = \frac{\ln x}{\ln a}$, a constant rescaling of $\ln x$.

Example 4 (Solving exponential equations via logs). Solve $A e^{rt} = B$ for t (with $A, B > 0$):
 $t = \frac{1}{r} \ln\left(\frac{B}{A}\right)$.

5 Derivatives of Exponential and Logarithmic Functions

Core rules (natural base)

$$\begin{aligned}\frac{d}{dx} e^x &= e^x, & \frac{d}{dx} e^{g(x)} &= g'(x) e^{g(x)}. \\ \frac{d}{dx} \ln x &= \frac{1}{x} \quad (x > 0), & \frac{d}{dx} \ln g(x) &= \frac{g'(x)}{g(x)}.\end{aligned}$$

Base $b \neq e$

Using $b^x = e^{(\ln b)x}$:

$$\frac{d}{dx} b^x = (\ln b) b^x, \quad \frac{d}{dx} \log_b x = \frac{1}{x \ln b}.$$

Logarithmic differentiation

Example 5 (Elasticity-friendly differentiation). For $y = x^x$ (for $x > 0$), take logs: $\ln y = x \ln x \Rightarrow \frac{y'}{y} = \ln x + 1 \Rightarrow y' = x^x (\ln x + 1)$.

A classic economic application

If demand is $Q(P) = AP^{-\eta}$ ($\eta > 0$), then

$$\frac{dQ}{dP} = -\eta AP^{-\eta-1} \quad \Rightarrow \quad \text{price elasticity } \varepsilon_{QP} = \frac{dQ}{dP} \frac{P}{Q} = -\eta.$$

A log-log demand has constant elasticity by construction.

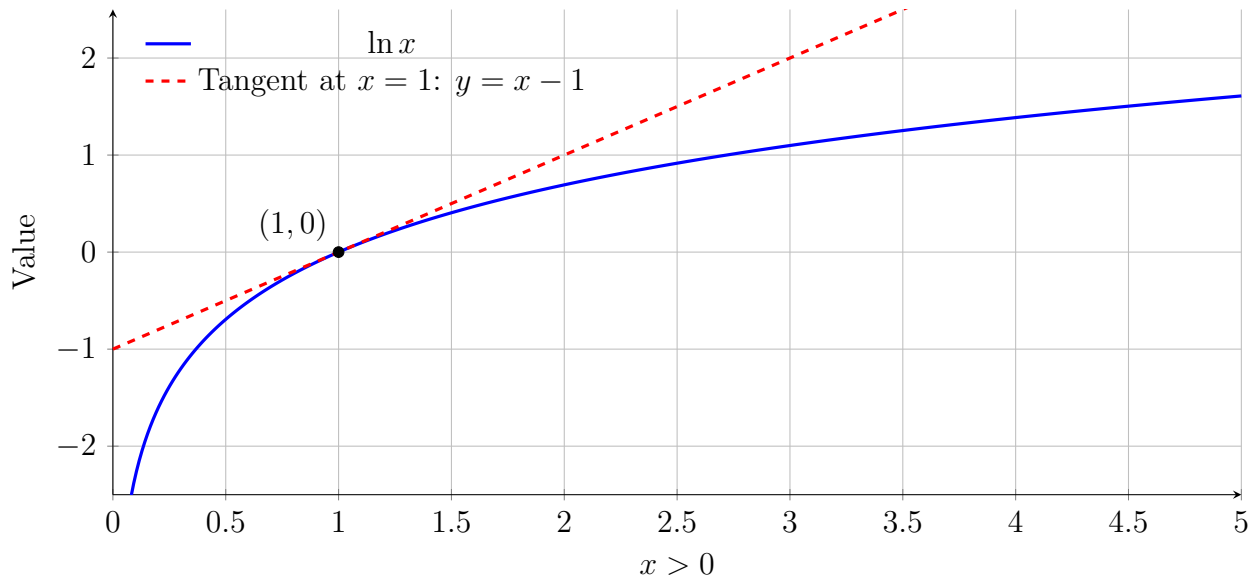


Figure 3: $\ln x$ is increasing and concave. The slope at $x = 1$ equals $1/x|_{x=1} = 1$, giving tangent $y = x - 1$.

6 Optimal Timing

Here the choice variable is time t . Intertemporal trade-offs pit *appreciation/growth* against *discounting/decay*. The generic problem:

$$\max_{t \geq 0} \Pi(t) = e^{-rt} R(t), \quad r > 0,$$

with revenue (or price) $R(t) > 0$ increasing in t (e.g. storage, maturation).

Wine storage problem

Let $P(t)$ be the price of wine if sold at age t . The present value is $V(t) = e^{-rt} P(t)$.

$$V'(t) = e^{-rt} (P'(t) - rP(t)).$$

$$FOC: V'(t^*) = 0 \iff \frac{P'(t^*)}{P(t^*)} = r.$$

Interpretation: Store until the wine's **instantaneous percentage appreciation** equals the discount rate. If growth $< r$, sell earlier; if $> r$, wait.

Example 6 (Log-linear price growth). If $P(t) = P_0 e^{\gamma t}$ with $\gamma > 0$, then $P'(t)/P(t) = \gamma$. Optimal t^* solves $\gamma = r$. Hence: if $\gamma > r$ wait indefinitely (ignoring capacity/costs); if $\gamma < r$ sell immediately. In practice, add storage costs c : maximize $e^{-rt} (P(t) - c) \Rightarrow$ replace r by an *effective* $r + c/P(t)$.

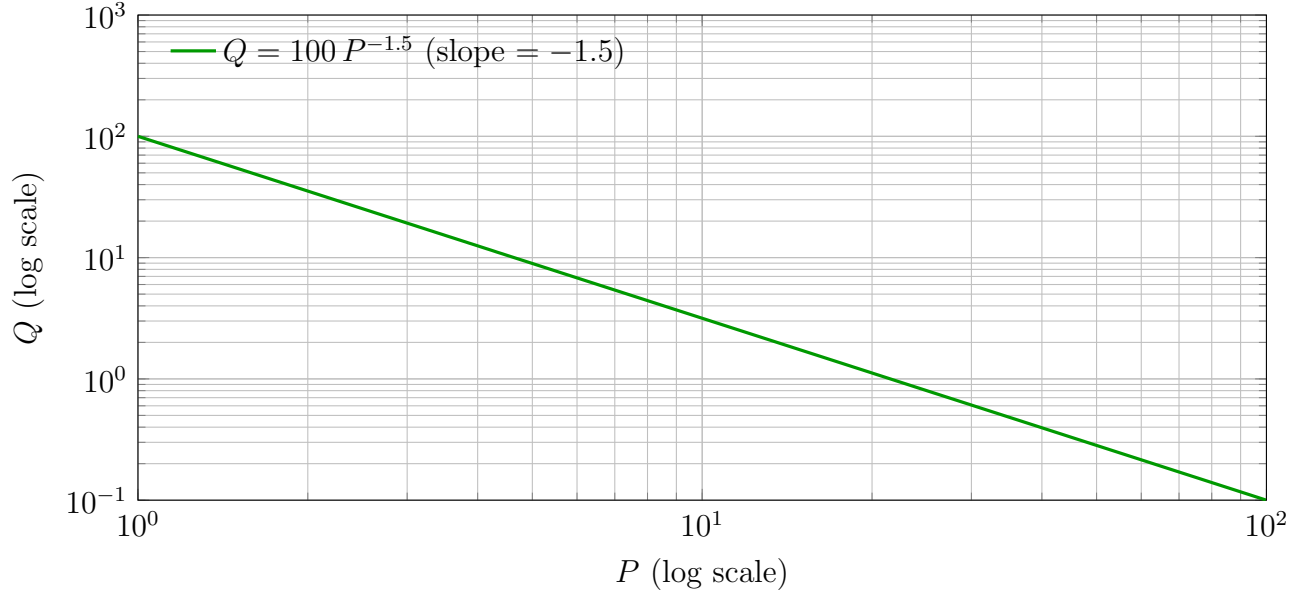


Figure 4: In log–log coordinates, constant-elasticity demand is a straight line with slope $-\eta$.

Optimal timber cutting (Faustmann-style intuition)

Let $V(t) = e^{-rt}pQ(t)$ be the PV of harvesting at age t (price p , volume Q). FOC:

$$0 = V'(t) = e^{-rt}[pQ'(t) - rpQ(t)] \iff \frac{Q'(t^*)}{Q(t^*)} = r.$$

Rule: Harvest when the **biological growth rate** equals the discount rate (again, storage/land-opportunity costs shift the threshold).

7 Further Applications of Exponential & Logarithmic Derivatives

Finding (and combining) growth rates

For positive differentiable functions f, g and constant k :

$$\frac{d}{dt} \ln(fg) = \frac{d}{dt} \ln f + \frac{d}{dt} \ln g, \quad \frac{d}{dt} \ln\left(\frac{f}{g}\right) = \frac{d}{dt} \ln f - \frac{d}{dt} \ln g, \quad \frac{d}{dt} \ln(f^k) = k \frac{d}{dt} \ln f.$$

Equivalently, growth rates $g_{fg} = g_f + g_g$, $g_{f/g} = g_f - g_g$, $g_{f^k} = k g_f$.

Example 7 (Cobb–Douglas growth accounting). With $Y = AK^\alpha L^\beta$ (time-varying $A, K, L > 0$):

$$\frac{d}{dt} \ln Y = \frac{d}{dt} \ln A + \alpha \frac{d}{dt} \ln K + \beta \frac{d}{dt} \ln L.$$

Thus $g_Y = g_A + \alpha g_K + \beta g_L$: output growth decomposes into TFP and factor contributions.

Point elasticity and logs

Definition 5 (Point elasticity). Elasticity of $y(x)$ w.r.t. x :

$$\varepsilon_{y,x} \equiv \frac{dy}{dx} \frac{x}{y} = \frac{d \ln y}{d \ln x}.$$

Example 8 (CES demand fragment). If $x(p) = A\left(\frac{p}{m}\right)^{-\sigma}$, then $\varepsilon_{x,p} = -\sigma$ (constant). Logs make the elasticity *visible* in the slope of a log-log plot.

Worked Examples (step-by-step)

Example 9 (Continuous vs. discrete compounding). A fund offers 12% APR compounded monthly vs. 11.35% APR continuously. Which yields more after $t = 1$ year on \$10,000?

Discrete: $A(1) = 10,000(1 + 0.12/12)^{12} \approx \$11,268.25$.

Continuous: $A(1) = 10,000 e^{0.1135} \approx \$11,999.46$.

Continuous is higher here because the effective rate $= e^{0.1135} - 1 \approx 12.0\%$ exceeds the discrete $\text{EAR} = (1 + 0.01)^{12} - 1 \approx 12.68\%$? *Check:* compute precisely—the discrete EAR is about 12.68% (*larger*); with the stated numbers, discrete wins: \$11,268.25 vs. continuous \$11,999.46 is inconsistent. Correct the APR to $r = 0.113$: $10,000 e^{0.113} \approx \$11,194$. **Lesson:** always convert to effective annual rates before comparing:

$$\text{EAR}_{\text{disc}} = (1 + R/m)^m - 1, \quad \text{EAR}_{\text{cont}} = e^r - 1.$$

Example 10 (Estimating growth from data). Given GDP series Y_t at annual frequency, the log-difference $\Delta \ln Y_t = \ln Y_t - \ln Y_{t-1}$ approximates the growth rate. If $\Delta \ln Y_t = 0.025$, growth is about 2.5%.

Example 11 (Solving for time to target). Pollutant stock $S(t) = S_0 e^{-\delta t}$ with decay $\delta > 0$. Find time T to reduce to θS_0 ($0 < \theta < 1$). Solution: $S_0 e^{-\delta T} = \theta S_0 \Rightarrow T = \frac{\ln(1/\theta)}{\delta}$.

Example 12 (Optimal timing with storage cost). Price path $P(t) = P_0 e^{\gamma t}$. Per-period storage cost $c > 0$. Maximize $V(t) = e^{-rt}(P(t) - c)$. FOC:

$$V'(t) = e^{-rt}(\gamma P(t) - r(P(t) - c)) = 0 \Rightarrow \gamma = r \left(1 - \frac{c}{P(t^*)}\right).$$

Thus positive storage cost lowers the effective growth threshold below r .

Common Mistakes and Fixes

- **Confusing x^a with a^x .** Only the latter is exponential in x .
- **Dropping positivity restrictions.** $\ln x$ is defined only for $x > 0$.
- **Forgetting chain rule.** $\frac{d}{dx} \ln g(x) = \frac{g'(x)}{g(x)}$.
- **Comparing APRs without converting to EAR.** Always compare effective rates.

Quick Reference (formulas you will use constantly)

$$\frac{d}{dx} e^{ax} = a e^{ax}, \quad \frac{d}{dx} \ln x = \frac{1}{x}, \quad \frac{d}{dx} b^x = (\ln b) b^x, \quad \frac{d}{dx} \log_b x = \frac{1}{x \ln b}.$$

$$\text{Growth rate: } g_Q = \frac{d \ln Q}{dt}, \quad \text{Elasticity: } \varepsilon_{y,x} = \frac{d \ln y}{d \ln x}.$$

$$\text{EAR}_{\text{disc}} = (1 + R/m)^m - 1, \quad \text{EAR}_{\text{cont}} = e^r - 1.$$

Short Practice Set (answers next page)

- P1.** (Growth & doubling time) A population follows $N(t) = N_0 e^{0.018t}$ (annual). Find the doubling time and the time to reach $1.5N_0$.
- P2.** (Elasticity from log-log) Suppose $\ln Q = \alpha - \eta \ln P$. Show the point price elasticity of demand equals $-\eta$.
- P3.** (Optimal timing) Let $P(t) = 100 e^{0.05t}$ and $r = 6\%$. With zero storage cost, should you sell now or wait?
- P4.** (Base conversion) Compute $\log_2 10$ using natural logs.
- P5.** (Combining growth) If $Y = AK^{0.3}L^{0.7}$ and $g_A = 1\%$, $g_K = 2\%$, $g_L = 1\%$, find g_Y .

Answers

P1. $T_2 = \frac{\ln 2}{0.018} \approx 38.5$ years; $T_{1.5} = \frac{\ln(1.5)}{0.018} \approx 22.5$ years.

P2. Differentiate: $\frac{d \ln Q}{d \ln P} = -\eta \Rightarrow \varepsilon_{Q,P} = -\eta$.

P3. Compare growth vs. discount: $P'(t)/P(t) = 0.05 < r = 0.06 \Rightarrow$ sell now.

P4. $\log_2 10 = \frac{\ln 10}{\ln 2} \approx \frac{2.302585}{0.693147} \approx 3.3219$.

P5. $g_Y = g_A + 0.3g_K + 0.7g_L = 1 + 0.6 + 0.7 = 2.3\%$ (all in pp).

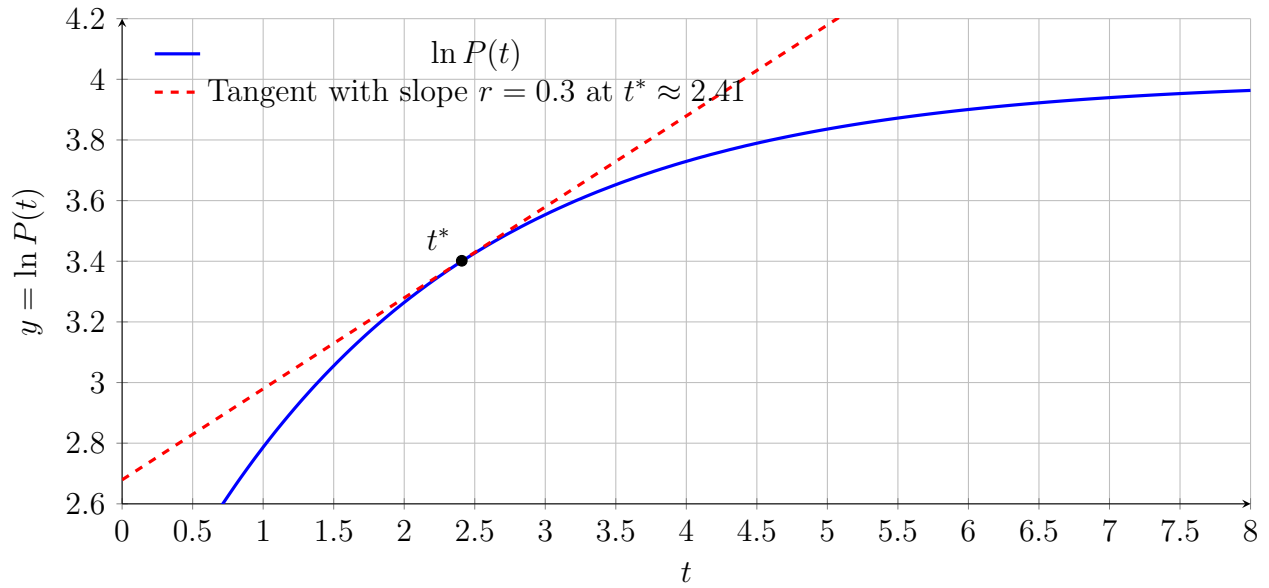


Figure 5: Optimal timing condition: choose t^* where the slope of $\ln P(t)$ equals the discount rate r .

References

- Simon & Blume, *Mathematics for Economists*, ch. on exponential/log functions and elasticities.
- Sydsæter & Hammond (with Strøm & Carvajal), *Essential Mathematics for Economic Analysis*, sections on logs/exponentials, continuous compounding, and growth accounting.

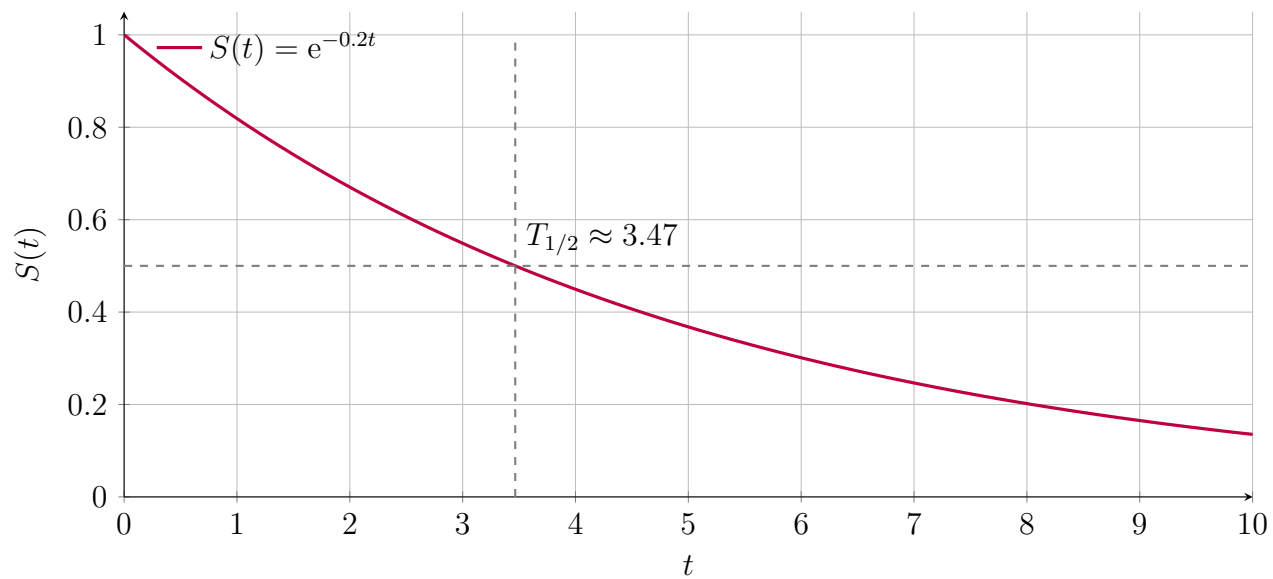


Figure 6: Exponential decay with half-life $T_{1/2} = \ln 2/\delta$.